DAY 42

Detecting Data Drift with Python

Comparing training and production data with pandas

Learning outcome: Compare feature summaries, calculate mean shifts, create a basic warning, and understand the limits of simple drift checks.

1. What is data drift?

Data drift occurs when production input features have a different distribution from a reference dataset such as the training data. For example, students in training data may study 1–5 hours per day, while recent production inputs increasingly report 6–9 hours.

2. Related concepts

Data drift: input feature distributions change.
Concept drift: the relationship between inputs and target changes.
Prediction drift: the distribution of model predictions changes.

A drift warning does not automatically mean model accuracy has declined. Investigate the change and evaluate predictions against actual outcomes when labels are available.

3. Compare two samples

```
training = pd.Series([1,2,2,3,3,4,4,5])
production = pd.Series([4,5,6,6,7,8,8,9])
training_mean = training.mean()       # 3.0
production_mean = production.mean()   # 6.625
difference = abs(training_mean - production_mean)  # 3.625
```

The absolute mean difference is 3.625 hours. This signals a shift in average study hours in this synthetic example.

4. Threshold-based warning

```
threshold = 2.0
if difference > threshold:
    print("Warning: Possible data drift detected!")
else:
    print("No mean-shift warning.")
```

This is an educational rule, not a formal statistical test. Choose thresholds using domain knowledge and normal variation. Do not automatically retrain a model just because a single threshold was crossed.

5. Why check standard deviation too?

The mean describes the average. Standard deviation describes spread. Two datasets can have similar means but different spreads, so a mean-only check may miss meaningful distribution changes.

```
print(training.std())
print(production.std())
```

6. More robust monitoring

Real projects can monitor missing-value rates, out-of-range values, quantiles, categorical frequencies, and feature distributions over time. Depending on the data, teams may use tests such as Kolmogorov–Smirnov for suitable continuous samples or chi-square for suitable categorical counts. Statistical tests have assumptions and must be interpreted in context.

7. Run the practice file

```
python -m pip install pandas numpy pytest\npython Day-42-Data-Drift-Detection-Practice.py\npytest -v Day-42-Data-Drift-Detection-
```

The Python file contains reusable functions and six tests for means, thresholds, empty data, and missing values. It uses synthetic data and requires no trained model artifact.

8. Practice tasks with solutions

Task 1: Training mean = 3.0 and production mean = 6.625. Difference? **Solution:** 3.625.
Task 2: Difference = 1.2 and threshold = 2.0. Warning? **Solution:** No.
Task 3: Why compare standard deviation? **Solution:** Spread can change even when the mean stays similar.
Task 4: Does drift prove accuracy fell? **Solution:** No, use actual labels to evaluate quality when available.

9. Day 42 checklist

- ■ I can explain data drift and concept drift.
- ■ I can compare means in pandas.
- ■ I can calculate absolute mean difference.
- ■ I understand thresholds are context-dependent.
- ■ I know mean comparison is not complete drift detection.

GitHub learning repository: https://github.com/DeveloperDesigner-star/Datasciencepdfs